

## QUEEN BEATRIX INTL, Aruba

WMO: 789820

Lat: 12.500N

Lon: 70.017W

Elev: 18

StdP: 101.11

Time Zone: -4.00 (W04)

Period: 94-19

WBAN: 00411

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB  |             | Humidification DP/MCDB and HR |             |             |             |             |             | Coldest Month WS/MCDB |             |             |             | MCWS/PCWD to 99.6% DB |           | WSF          |
|---------------|-------------|-------------|-------------------------------|-------------|-------------|-------------|-------------|-------------|-----------------------|-------------|-------------|-------------|-----------------------|-----------|--------------|
|               |             |             | 99.6%                         |             |             | 99%         |             |             | 0.4%                  |             | 1%          |             |                       |           |              |
|               | 99.6%       | 99%         | DP                            | HR          | MCDB        | DP          | HR          | MCDB        | WS                    | MCDB        | WS          | MCDB        | MCWS                  | PCWD      |              |
| (a)<br>1      | (b)<br>23.9 | (c)<br>24.4 | (d)<br>19.9                   | (e)<br>14.6 | (f)<br>26.3 | (g)<br>20.4 | (h)<br>15.1 | (i)<br>26.4 | (j)<br>12.4           | (k)<br>29.3 | (l)<br>11.8 | (m)<br>29.1 | (n)<br>4.4            | (o)<br>80 | (p)<br>0.733 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 9             | 5.5                    | 33.9            | 27.2 | 33.2 | 27.0 | 33.0 | 26.9 | 28.2                | 32.1 | 27.8 | 31.7 | 27.5 | 31.4 | 8.7                  | 80   |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 27.1                            | 22.9 | 31.1 | 26.8 | 22.5 | 30.8 | 26.2 | 21.7 | 30.3 | 91.0          | 32.1 | 88.8 | 31.9 | 87.4 | 31.8 | 32.8              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |      |                    | n-Year Return Period Values of Extreme Temperature |           |      |            |      |            |      |            |      |      |
|-------------------|------|------|----------------------------|------|------|--------------------|----------------------------------------------------|-----------|------|------------|------|------------|------|------------|------|------|
| 1%                |      | 2.5% | 5%                         | Mean |      | Standard Deviation |                                                    | n=5 years |      | n=10 years |      | n=20 years |      | n=50 years |      |      |
| Min               |      | Max  |                            | Min  | Max  |                    | Min                                                | Max       | Min  | Max        | Min  | Max        | Min  | Max        |      |      |
| (a)               | (b)  | (c)  |                            | (e)  | (f)  | (g)                | (h)                                                | (i)       | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |      |
| 12.3              | 11.6 | 11.1 |                            | DB   | 22.3 | 35.2               | 0.7                                                | 0.6       | 21.8 | 35.6       | 21.4 | 36.0       | 21.0 | 36.4       | 20.5 | 36.8 |
|                   |      |      |                            | WB   | 20.5 | 29.2               | 0.7                                                | 1.2       | 20.0 | 30.0       | 19.6 | 30.8       | 19.2 | 31.4       | 18.7 | 32.3 |

## Monthly Climatic Design Conditions

|                                                                               |                                                                               | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                               |                                                                               | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                           | DBAvg                                                                         | 29.0   | 27.5  | 27.7  | 28.2  | 28.9  | 29.4  | 29.7  | 29.6  | 30.2  | 30.3  | 29.5  | 28.9  | 28.0 |
|                                                                               | DBStd                                                                         | 1.25   | 0.81  | 0.73  | 0.79  | 0.76  | 0.74  | 0.70  | 0.70  | 0.73  | 0.96  | 1.13  | 1.06  | 0.95 |
|                                                                               | HDD10.0                                                                       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    |
|                                                                               | HDD18.3                                                                       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    |
|                                                                               | CDD10.0                                                                       | 6932   | 543   | 495   | 563   | 567   | 602   | 591   | 607   | 626   | 609   | 606   | 566   | 558  |
|                                                                               | CDD18.3                                                                       | 3891   | 285   | 262   | 304   | 317   | 344   | 341   | 348   | 368   | 359   | 347   | 316   | 299  |
|                                                                               | CDH23.3                                                                       | 45301  | 2782  | 2584  | 3160  | 3561  | 4104  | 4181  | 4253  | 4679  | 4672  | 4364  | 3754  | 3205 |
|                                                                               | CDH26.7                                                                       | 17564  | 682   | 682   | 939   | 1252  | 1648  | 1789  | 1787  | 2204  | 2286  | 1924  | 1437  | 933  |
| Wind                                                                          | WSAvg                                                                         | 7.4    | 7.1   | 7.8   | 8.0   | 8.0   | 8.3   | 8.8   | 8.3   | 7.9   | 6.8   | 5.7   | 5.6   | 6.7  |
| Precipitation                                                                 | PrecAvg                                                                       | 417    | 43    | 21    | 10    | 15    | 13    | 22    | 31    | 17    | 37    | 55    | 87    | 64   |
|                                                                               | PrecMax                                                                       | 624    | 106   | 79    | 36    | 109   | 67    | 102   | 118   | 50    | 121   | 164   | 223   | 205  |
|                                                                               | PrecMin                                                                       | 197    | 14    | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 3     | 4     | 16    | 8    |
|                                                                               | PrecStd                                                                       | 137    | 27    | 21    | 10    | 22    | 15    | 28    | 29    | 14    | 33    | 41    | 61    | 53   |
| Monthly Design<br>Dry Bulb and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%                                                                          | DB     | 31.2  | 31.6  | 32.2  | 33.0  | 33.3  | 33.9  | 33.3  | 34.2  | 34.5  | 34.0  | 33.0  | 32.1 |
|                                                                               |                                                                               | MCWB   | 25.3  | 25.9  | 26.2  | 26.8  | 27.2  | 27.9  | 26.7  | 26.9  | 27.2  | 27.1  | 26.7  | 26.0 |
|                                                                               | 2%                                                                            | DB     | 30.8  | 31.0  | 31.4  | 32.2  | 32.9  | 33.0  | 33.0  | 33.6  | 34.0  | 33.1  | 32.2  | 31.2 |
|                                                                               |                                                                               | MCWB   | 25.2  | 25.4  | 25.7  | 26.3  | 27.1  | 27.3  | 26.7  | 26.9  | 27.3  | 26.9  | 26.5  | 25.7 |
|                                                                               | 5%                                                                            | DB     | 30.1  | 30.2  | 31.0  | 31.8  | 32.2  | 32.2  | 32.2  | 33.1  | 33.2  | 32.7  | 31.9  | 30.8 |
|                                                                               |                                                                               | MCWB   | 25.0  | 25.0  | 25.5  | 26.2  | 26.7  | 26.8  | 26.6  | 26.8  | 27.1  | 26.8  | 26.4  | 25.6 |
|                                                                               | 10%                                                                           | DB     | 29.8  | 30.0  | 30.3  | 31.1  | 31.8  | 31.9  | 31.9  | 32.6  | 32.9  | 32.0  | 31.1  | 30.1 |
|                                                                               |                                                                               | MCWB   | 24.9  | 24.8  | 25.1  | 26.0  | 26.5  | 26.7  | 26.5  | 26.6  | 27.0  | 26.7  | 26.4  | 25.4 |
|                                                                               | Monthly Design<br>Wet Bulb and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%   | WB    | 26.7  | 27.0  | 27.4  | 27.7  | 28.7  | 28.8  | 28.2  | 28.6  | 28.2  | 27.8  | 26.7 |
|                                                                               |                                                                               |        | MCDB  | 29.3  | 30.2  | 31.6  | 31.1  | 32.4  | 32.6  | 31.7  | 32.5  | 33.0  | 32.0  | 31.1 |
| 2%                                                                            |                                                                               | WB     | 26.0  | 26.1  | 26.5  | 27.1  | 27.7  | 27.9  | 27.6  | 27.6  | 28.0  | 27.7  | 27.2  | 26.4 |
|                                                                               |                                                                               | MCDB   | 29.5  | 29.8  | 30.6  | 31.1  | 31.3  | 31.9  | 31.3  | 32.0  | 32.3  | 31.4  | 30.8  | 29.9 |
| 5%                                                                            |                                                                               | WB     | 25.6  | 25.6  | 26.0  | 26.6  | 27.2  | 27.4  | 27.2  | 27.2  | 27.6  | 27.3  | 26.8  | 26.1 |
|                                                                               |                                                                               | MCDB   | 29.1  | 29.4  | 29.9  | 30.6  | 31.0  | 31.3  | 31.0  | 31.6  | 31.8  | 31.0  | 30.4  | 29.6 |
| 10%                                                                           |                                                                               | WB     | 25.2  | 25.2  | 25.6  | 26.3  | 26.7  | 26.9  | 26.7  | 26.9  | 27.2  | 27.0  | 26.5  | 25.7 |
|                                                                               |                                                                               | MCDB   | 28.7  | 29.0  | 29.5  | 30.1  | 30.5  | 30.7  | 30.5  | 31.2  | 31.3  | 30.7  | 30.0  | 29.2 |
| Mean Daily<br>Temperature<br>Range                                            | 5% DB                                                                         | MDBR   | 5.0   | 5.3   | 5.3   | 5.3   | 5.2   | 5.2   | 5.3   | 5.5   | 5.5   | 5.1   | 4.9   | 4.8  |
|                                                                               |                                                                               | MCDBR  | 5.2   | 5.5   | 5.6   | 5.6   | 5.5   | 5.4   | 5.5   | 5.8   | 5.8   | 5.5   | 5.2   | 5.0  |
|                                                                               |                                                                               | MCWBR  | 2.5   | 2.6   | 2.6   | 2.4   | 2.4   | 2.4   | 2.3   | 2.2   | 2.3   | 2.3   | 2.3   | 2.3  |
|                                                                               | 5% WB                                                                         | MCDBR  | 5.0   | 5.3   | 5.3   | 5.4   | 5.2   | 5.4   | 5.3   | 5.5   | 5.6   | 5.2   | 5.0   | 4.8  |
|                                                                               |                                                                               | MCWBR  | 2.8   | 2.9   | 2.8   | 2.7   | 2.7   | 2.7   | 2.5   | 2.3   | 2.6   | 2.5   | 2.5   | 2.5  |
| Clear-Sky<br>Solar<br>Irradiance                                              | taub                                                                          | 0.388  | 0.412 | 0.421 | 0.486 | 0.475 | 0.497 | 0.479 | 0.444 | 0.428 | 0.402 | 0.393 | 0.394 |      |
|                                                                               | taud                                                                          | 2.542  | 2.447 | 2.435 | 2.233 | 2.301 | 2.229 | 2.291 | 2.426 | 2.454 | 2.561 | 2.576 | 2.553 |      |
|                                                                               | Ebn at Noon                                                                   | 915    | 903   | 898   | 833   | 828   | 800   | 817   | 857   | 878   | 901   | 903   | 900   |      |
|                                                                               | Edh at Noon                                                                   | 102    | 116   | 119   | 145   | 133   | 141   | 133   | 118   | 115   | 102   | 98    | 99    |      |
| All-Sky Solar<br>Radiation                                                    | RadAvg                                                                        | 5.33   | 5.97  | 6.29  | 6.38  | 6.27  | 6.39  | 6.53  | 6.68  | 6.28  | 5.51  | 5.03  | 4.93  |      |
|                                                                               | RadStd                                                                        | 0.21   | 0.16  | 0.25  | 0.20  | 0.25  | 0.22  | 0.17  | 0.21  | 0.36  | 0.34  | 0.49  | 0.26  |      |

## Historical Trends

|                        | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                        |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                        |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)     | (k)     | (l)     |
| Station Only           | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| Regional (2 neighbors) | N/A   | N/A     | -0.33  | -0.24   | -0.48 | -0.48 | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page